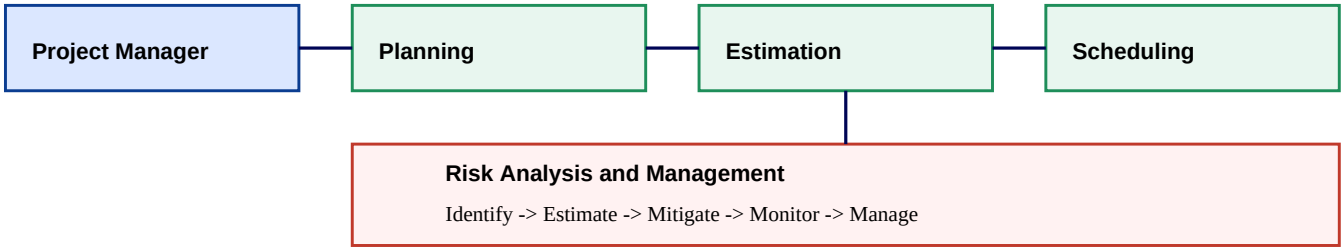
Software Engineering

Unit 5
Software Project Management

Complete, Easy-to-Understand, Exam-Ready Study Notes

DU-2304CS304 | Computer Science & Engineering

Based on: unit-5.pptx | Prepared for: Tanish Uchhadiya



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1. Software Project Manager

Simple definition

A **Software Project Manager (SPM)** is a person who plans, organizes, controls, and completes a software project successfully within the planned time, cost, scope, and quality.

Detailed explanation

The project manager is responsible for the complete project. The manager decides what work is required, divides the work into tasks, assigns tasks to team members, checks progress, solves problems, controls money, and ensures that the software is delivered on time.

For example, if a company wants to develop a food-delivery application, the project manager decides the features, assigns login work to one developer, restaurant-listing work to another, payment work to another, and testing work to testers. The manager then checks whether everyone is completing their work on time.

Key responsibilities

- **Project planning:** Prepare a roadmap for the project.
- **Objective setting:** Define what the project must achieve.
- **Task delegation:** Divide work and assign it to suitable people.
- **Resource management:** Manage people, tools, hardware, software, and money.
- **Risk management:** Identify possible problems and prepare solutions.
- **Problem solving:** Identify and resolve issues during the project.
- **Decision making:** Choose the best option when problems occur.
- **Progress tracking:** Compare actual work with the planned schedule.
- **Quality assurance:** Ensure that software meets requirements.
- **Budget control:** Keep project expenses within the planned budget.
- **On-time delivery:** Complete and deliver the software on schedule.
- **Communication:** Communicate with the team, customer, and senior management.
- **Team motivation:** Encourage team members and maintain teamwork.
- **Conflict resolution:** Solve disagreements within the team.
- **Success measurement:** Measure success using completed milestones and deliverables.

Important exam points

A project manager controls **scope, time, cost, quality, resources, communication, and risks**. Project management is successful when software is completed within the planned time and budget while meeting customer requirements.

2. Software Project Planning

Simple definition

Software project planning means preparing a clear roadmap before starting a software project. It helps estimate resources, cost, effort, schedule, and possible risks.

Detailed explanation

Before development begins, the project manager must know what software will be developed, whether it is possible, which people and tools are required, how much money is needed, and how much time the project will take. This information is written in the project plan.

A good project plan considers both the **best-case** and **worst-case** situations. In the best case, the project may finish on time without problems. In the worst case, requirements may change, staff may leave, or the project may take more time and money.

Main planning activities

Activity	Explanation
Define project scope	Decide which features and work are included in the project and which are not included.
Determine feasibility	Check whether the project is technically, financially, and practically possible.
Analyze risks	Identify possible problems before development begins.
Define resources	Identify people, reusable software, hardware, tools, and environment required.
Estimate cost and effort	Calculate the money and human work required.
Develop schedule	Decide the start and finish dates of every task.
Update plan	Change the plan when requirements, risks, or project conditions change.

Types of resources

Resource type	Examples
Human resources	Project manager, developers, testers, designers, analysts.
Reusable software resources	Existing code, libraries, frameworks, components, templates.
Environmental resources	Computers, servers, software tools, network, office space.

Important exam points

Project planning is not a one-time activity. It must be reviewed and updated throughout the project because requirements and conditions can change.

3. Software Project Estimation

Simple definition

Software project estimation is the process of predicting the effort, time, cost, and resources required to complete a software project.

Detailed explanation

Before starting a project, the manager needs to know approximately how much work is required, how long it will take, how much it will cost, and how many people or tools are needed. This process is called estimation.

Estimation depends on project size, complexity, number of features, team experience, technology, customer requirements, and availability of old project data.

Key components

Component	Meaning
Time	Duration required to complete the project.
Cost	Money required for salaries, tools, hardware, and other resources.
Resources	People, tools, hardware, software, and reusable components.
Effort	Total human work required, usually measured in person-months.

Methods of estimation

Method	Explanation	Example
Delay estimation	Wait until more project details are available.	If requirements are unclear, estimate after design.
Similar-project estimation	Use data from completed similar projects.	Use time and cost data from an old banking app.
Decomposition	Break the project into small parts and estimate each part.	Estimate login, payment, search, and reports separately.
Mathematical models	Use formulas and historical data.	COCOMO uses KLOC to estimate effort.

Important exam points

Effort is measured in **person-months (PM)**. One person-month means the work done by one person in one month.

4. Software Project Decomposition

Simple definition

Software project decomposition is the process of breaking a large software project into smaller, manageable tasks or modules.

Detailed explanation

A large software project is difficult to understand and manage. Therefore, it is divided into smaller parts. Each part can be planned, developed, tested, and tracked separately.

For example, a food-delivery application can be divided into user login, restaurant listing, menu browsing, order placement, payment, and order tracking. Each module can be given to a different developer or team.

Benefits

- Makes a large project easier to understand.
- Makes planning and development easier.
- Helps estimate effort, cost, time, and resources.
- Helps assign tasks to team members.
- Makes progress tracking easier.
- Reduces project complexity.
- Provides an organized development structure.

Decomposition techniques

- Software sizing.
- LOC-based estimation.
- Function Point estimation.
- Use-case-based estimation.
- Project-based estimation.

Important exam points

Decomposition is mainly used to make a large project simple and to improve the accuracy of estimation.

5. Software Sizing

Simple definition

Software sizing means measuring the size or amount of software that will be developed.

Detailed explanation

Software size is important because effort, time, cost, and staffing depend on it. If the size estimate is wrong, the complete project estimate may also be wrong.

Software size can be measured directly using **LOC** or indirectly using **Function Points**.

Methods of sizing

Method	Type	Meaning
LOC	Direct	Counts the number of lines of source code.
Function Point (FP)	Indirect	Measures the functionality delivered to users.

Factors affecting estimation accuracy

- Correct estimation of software size.
- Ability to convert size into effort, time, and cost.
- Experience and ability of the software team.
- Stability of customer requirements.
- Quality of the development environment.
- Type, domain, and complexity of the project.
- Number of components and features.
- Amount of existing software to be modified.

Important exam points

LOC measures code written; Function Points measure functionality delivered to users.

6. LOC-Based Estimation

Simple definition

LOC means **Lines of Code**. LOC-based estimation calculates project effort by estimating the total number of source-code lines.

Detailed explanation

In this method, the manager first estimates how many lines of code the software will need. Then, using the team's productivity rate, the manager calculates effort, cost, and schedule.

Formulas

Effort = Estimated LOC / Productivity
Cost = Effort x Cost per person-month
Schedule = Effort / Number of developers

Explanation of symbols

- **LOC:** Total estimated lines of source code.
- **Productivity:** Lines of code one person can write in one month.
- **Effort:** Total work required, measured in person-months.
- **Cost per person-month:** Salary or cost of one person for one month.
- **Schedule:** Time required to complete the project.

Solved example

Given information	Value
Estimated software size	25,000 LOC
Productivity	600 LOC per person-month
Team size	5 developers
Cost per person-month	$6,000

Step 1: Calculate effort.

Effort = 25,000 / 600 = 41.67, approximately 42 person-months

Step 2: Calculate schedule.

Schedule = 42 / 5 = 8.4 months

Step 3: Calculate cost.

Cost = 42 x 6,000 = $252,000

Advantages and limitations

Advantages	Limitations
Simple and easy to calculate.	Not suitable in early stages.
Useful when design and code are known.	Difficult for new technology.
Measures developer productivity.	More code does not always mean more functionality.

7. Function Point Metrics

Simple definition

A **Function Point (FP)** measures the functionality delivered by software to the user. It does not directly depend on the programming language or number of code lines.

Detailed explanation

Function Points measure what the user can do in the software. For example, login, searching, generating reports, storing data, and using external data are functionalities.

Uses of Function Points

- Estimate project cost and effort.
- Estimate development time.
- Predict the number of errors during testing.
- Estimate the number of software components.
- Estimate approximate source-code lines.

Five Function Point components

Component	Full form	Meaning	Example
EI	External Input	Data enters the system from a user or another application.	Login form, registration form, payment details.
EO	External Output	Software provides processed information to the user.	Reports, receipts, error messages.
EQ	External Inquiry	Input produces an immediate software response.	Search a product and see details.
ILF	Internal Logical File	Data is stored and maintained inside the application.	User, product, or order database.
EIF	External Interface File	Software uses data stored outside the application.	Bank data used by a payment app.

Function Point formula

FP = Count Total x [0.65 + 0.01 x Sum(Fi)]

Explanation of symbols

- **Count Total:** Total of all FP entries.
- **Fi:** Value adjustment factors; there are 14 factors.
- **Sum(Fi):** Total value of all adjustment factors.
- **0.65:** Base value in the formula.
- **0.01:** Multiplying factor for adjustment values.

Solved example

If Count Total = 50 and Sum of adjustment factors = 17:

FP = 50 x [0.65 + 0.01 x 17] = 50 x 0.82 = 41 Function Points

Important exam points

LOC measures how much code is written. Function Points measure how much functionality is delivered to users.

8. COCOMO Model

Simple definition

COCOMO means **Constructive Cost Estimation Model**. It was proposed by **Barry Boehm** and is used to estimate software effort, development time, cost, and team size.

Detailed explanation

COCOMO uses the estimated size of software, usually in KLOC, to calculate effort and development time. The constants used in the formula depend on the type of software project.

Important terms

Term	Meaning
KLOC	Kilo Lines of Code. 1 KLOC = 1,000 lines of code.
Effort	Total work required, measured in person-months.
Tdev	Estimated development time in months.
Person-month	Work done by one person in one month.

Basic COCOMO formulas

Effort = a1 x (KLOC)^a2
Development Time = b1 x (Effort)^b2

Types of software projects

Project type	Characteristics	Example
Organic	Small, simple, experienced team, stable requirements.	Small business or college application.
Semidetached	Medium size, mixed team experience, medium complexity.	Medium banking or inventory system.
Embedded	Large, complex, strict hardware and software constraints.	Aircraft control or medical software.

COCOMO constants

Project type	a1	a2	b1	b2
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Three COCOMO models

- **Basic COCOMO:** Uses KLOC and project type.
- **Intermediate COCOMO:** Uses KLOC, project type, and cost drivers.
- **Complete COCOMO:** Considers every phase of the software life cycle separately.

Important exam points

100 person-months does not always mean 100 people working for one month or one person working for 100 months. It represents total work; actual duration depends on task dependencies and team size.

9. Project Scheduling and Tracking

Simple definition

Project scheduling means distributing estimated effort across the planned project duration and assigning tasks to team members with start and finish dates.

Detailed explanation

Scheduling decides what work will be done, who will do it, when it will start, and when it will finish. It also identifies which tasks can be done together and which tasks must wait for other tasks.

Scheduling principles

Principle	Meaning
Compartmentalization	Divide the project into manageable tasks.
Interdependency	Identify parallel and serial tasks.
Time allocation	Give start and finish dates to tasks.
Effort validation	Ensure enough staff are available.
Define responsibilities	Assign every task to a person.
Define outcomes	Every task must produce a deliverable.
Define milestones	Important completed achievements after quality review.

40-20-40 rule

Activity	Effort
Analysis and design	40% or more
Programming	20%
Testing	40%

Types of software projects

Project type	Meaning
Concept development	A new business idea or technology is explored.
New application development	A completely new software product is made.
Application enhancement	Existing software is improved.
Application maintenance	Existing software is corrected or updated.
Reengineering	Old software is rebuilt or modernized.

10. PERT, CPM and Gantt Chart

PERT

PERT means **Program Evaluation and Review Technique**. It is used when task time is uncertain. It helps estimate the most likely time for each task and the total project duration.

CPM

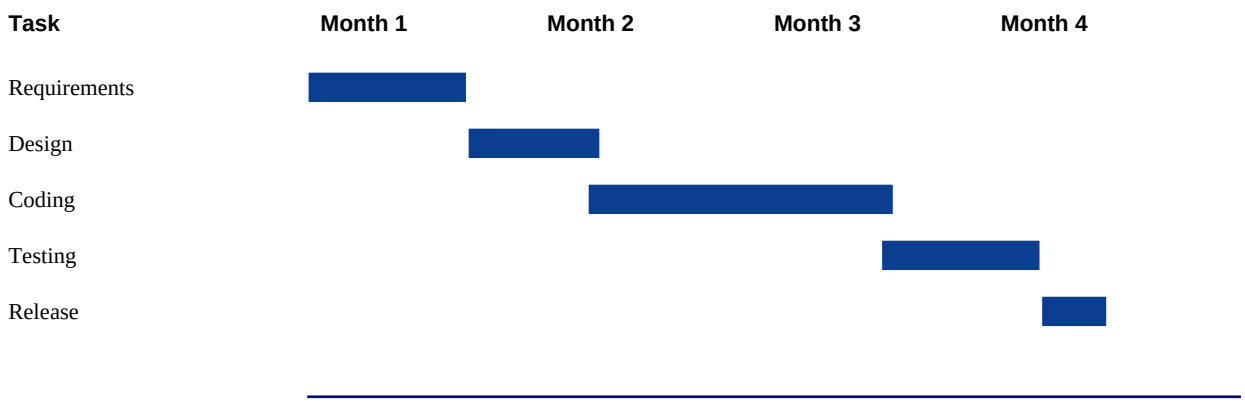
CPM means **Critical Path Method**. It helps find the critical path and identify task dependencies.

Critical path

The critical path is the longest chain of dependent tasks. It decides the minimum time required to complete the project. If a critical-path task is delayed, the whole project may be delayed.

Gantt chart

A **Gantt chart** shows project tasks against time. Activities are listed on the left, time is shown at the top, and each task is represented by a horizontal bar.



Gantt chart shows

- What tasks must be completed.
- When each task starts and ends.
- How long each task takes.
- Which tasks overlap.
- The total project start and end date.

Schedule tracking methods

- Conduct regular project status meetings.
- Check whether milestones are completed.
- Compare actual dates with planned dates.
- Review development and testing results.
- Use Earned Value Analysis for progress measurement.
- Monitor staff, resources, problems, reviews, and budget.

11. Risk Analysis and Management

Simple definition

A **risk** is a possible future problem that may or may not happen. If it happens, it may cause delay, extra cost, or loss.

Detailed explanation

Risk has two main characteristics: uncertainty and loss. Uncertainty means the risk may or may not happen. Loss means that if the risk occurs, it can damage the project or product.

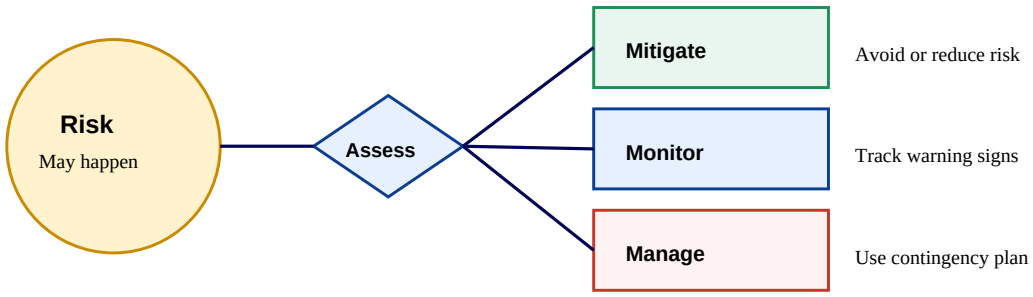
For example, a developer may leave the project, the customer may change requirements, or the project may take more time than expected.

Types of risks

Risk type	What it affects	Possible result
Project risk	Project plan	Schedule delay, increased cost, lack of staff or resources.
Technical risk	Quality and implementation	Poor performance or difficult implementation.
Business risk	Product feasibility	Product may not be useful or may not support business goals.

Business risk categories

Risk	Meaning
Market risk	Software is good but nobody wants to use it.
Strategic risk	Software does not match company strategy.
Sales risk	Sales team cannot sell the software.
Management risk	Senior management stops supporting the project.
Budget risk	Money or staff commitment is reduced.



Risk management steps

- **Risk identification:** Find possible risks.
- **Risk analysis:** Estimate probability and impact.
- **Risk ranking:** Give priority to serious risks.
- **Contingency planning:** Prepare an emergency plan.

Risk checklist categories

- **Product size:** Risks due to large software size.
- **Business impact:** Risks due to management or market constraints.

- **Customer characteristics:** Risks due to poor communication or inexperienced customer.
- **Process definition:** Risks due to undefined development process.
- **Development environment:** Risks due to poor tools or environment.
- **Technology:** Risks due to complex or new technology.
- **Staff size and experience:** Risks due to small or inexperienced team.

12. Risk Estimation, RMMM and RIS

Risk estimation

Risk estimation, also called risk projection, evaluates each risk using **probability** and **impact**. Probability means how likely the risk is to occur. Impact means how much damage it can cause.

Risk estimation steps

- Create a probability scale, for example 1 = low and 10 = high.
- Explain the possible effect of the risk.
- Estimate the impact on schedule, cost, quality, and customer satisfaction.
- Record the result clearly.
- Give highest priority to risks with high probability and high impact.

RMMM

RMMM means **Risk Mitigation, Monitoring, and Management**. It is a strategy to handle risks.

Part	Meaning	Example
Risk mitigation	Avoid or reduce the risk before it occurs.	Train another developer for an important developer's work.
Risk monitoring	Track warning signs and check whether risk is increasing.	Check employee satisfaction and work progress.
Risk management	Use contingency plan after risk occurs.	Assign trained replacement developer if main developer leaves.

RMMM Plan

An **RMMM Plan** documents all work performed during risk analysis and becomes part of the overall project plan.

RIS

RIS means **Risk Information Sheet**. It records complete information about one risk.

RIS contains

- Risk description.
- Risk category.
- Probability of occurrence.
- Impact and priority.
- Mitigation plan.
- Monitoring method.
- Contingency plan.
- Person responsible for the risk.

RIS is often maintained in a database so risks can be searched, updated, prioritized, and analyzed easily.

13. Exam Practice Questions

Short-answer questions

Q1. What is software project management?

Answer: Software project management is the process of planning, organizing, controlling, and completing a software project successfully within time, cost, scope, and quality.

Q2. What is LOC?

Answer: LOC means Lines of Code. It is a direct method of measuring software size by counting source-code lines.

Q3. What is a Function Point?

Answer: A Function Point measures the functionality delivered by software to the user.

Q4. What is COCOMO?

Answer: COCOMO means Constructive Cost Estimation Model. It estimates software effort, development time, cost, and team size using KLOC.

Q5. What is a milestone?

Answer: A milestone is an important completed achievement in a project. It is complete when its work product passes quality review.

Q6. What is a risk?

Answer: A risk is a possible future problem that may or may not happen and may cause loss if it occurs.

Q7. What is RMMM?

Answer: RMMM means Risk Mitigation, Monitoring, and Management.

Q8. What is a Gantt chart?

Answer: A Gantt chart is a project management tool that displays tasks against time using horizontal bars.

Long-answer questions

Q1. Explain the responsibilities of a software project manager.

Answer: A software project manager plans the project, sets objectives, divides work, assigns tasks, manages resources, controls budget, manages risks, solves problems, tracks progress, ensures quality, communicates with stakeholders, motivates the team, resolves conflicts, and ensures on-time delivery.

Q2. Explain LOC-based estimation with an example.

Answer: LOC-based estimation calculates effort using estimated lines of code and productivity. Effort = LOC / Productivity. For example, if estimated LOC is 25,000 and productivity is 600 LOC per person-month, effort is approximately 42 person-months. With 5 developers, schedule is about 8.4 months. If cost per person-month is $6,000, total cost is $252,000.

Q3. Explain Function Point metrics and its components.

Answer: Function Points measure software functionality. The five components are External Inputs, External Outputs, External Inquiries, Internal Logical Files, and External Interface Files. FP is calculated using: FP = Count Total x [0.65 + 0.01 x Sum(Fi)].

Q4. Explain COCOMO model and its project types.

Answer: COCOMO is a Constructive Cost Estimation Model proposed by Barry Boehm. It estimates effort and development time using KLOC. The three project types are Organic, Semidetached, and Embedded. Organic projects are small and simple; semidetached projects are medium; embedded projects are large and complex.

MCQs

What does LOC stand for?

A. A. Line of Control
B. B. Lines of Code
C. C. Level of Complexity
D. D. Length of Code

Correct answer: B

LOC means Lines of Code.

Which model was proposed by Barry Boehm?
A. A. COCOMO
B. B. PERT
C. C. CPM
D. D. Gantt
Correct answer: A
COCOMO was proposed by Barry Boehm.

Which method measures functionality delivered to users?
A. A. LOC
B. B. Function Point
C. C. KLOC
D. D. CPM
Correct answer: B
Function Points measure functionality.

What is the effort distribution in the 40-20-40 rule?
A. A. 40% design, 20% coding, 40% testing
B. B. 20% design, 40% coding, 40% testing
C. C. 40% coding, 40% testing, 20% design
D. D. 50% design, 25% coding, 25% testing
Correct answer: A
40-20-40 means design, coding, testing.

Which chart shows tasks against time?
A. A. Pie chart
B. B. Gantt chart
C. C. Histogram
D. D. Flowchart
Correct answer: B
Gantt chart shows tasks against time.

What does RMMM stand for?
A. A. Risk Mitigation, Monitoring, and Management
B. B. Risk Measurement and Model Management
C. C. Resource Management and Maintenance
D. D. Risk Management Method Model
Correct answer: A
RMMM means Risk Mitigation, Monitoring, and Management.

Which path determines the minimum project duration?
A. A. Shortest path
B. B. Critical path
C. C. Random path
D. D. Parallel path
Correct answer: B
Critical path determines project duration.

Which is an example of an embedded project?

A. A. College result system

B. B. Aircraft control software

C. C. Small billing application

D. D. Personal blog

Correct answer: B

Aircraft control software is embedded.

14. Final Revision Section

Chapter-wise revision

Topic	Remember
Project manager	Controls time, cost, quality, resources, and risks.
Planning	Creates roadmap and estimates resources, cost, effort, schedule.
Estimation	Predicts effort, time, cost, and resources.
Decomposition	Breaks large project into smaller tasks.
LOC	Measures lines of code.
Function Point	Measures user functionality.
COCOMO	Estimates effort and time using KLOC.
Scheduling	Assigns tasks, people, and dates.
40-20-40 rule	40% design, 20% coding, 40% testing.
PERT	Estimates project duration.
CPM	Finds critical path.
Gantt chart	Shows tasks against time.
Risk	Possible future problem.
RMMM	Mitigation, Monitoring, and Management.
RIS	Risk Information Sheet.

Important definitions

- **Software Project Manager:** Person responsible for completing a software project successfully.
- **Project Planning:** Preparing a roadmap for the software project.
- **Estimation:** Predicting effort, time, cost, and resources.
- **Decomposition:** Breaking a large project into smaller tasks.
- **LOC:** Lines of Code.
- **Function Point:** Measure of software functionality.
- **COCOMO:** Constructive Cost Estimation Model.
- **Milestone:** Important completed achievement.
- **Risk:** Possible future problem.
- **RMMM:** Risk Mitigation, Monitoring, and Management.
- **RIS:** Risk Information Sheet.

Important formulas

Effort = LOC / Productivity
Cost = Effort x Cost per person-month
Schedule = Effort / Number of developers
FP = Count Total x [0.65 + 0.01 x Sum(Fi)]

Effort = a1 x (KLOC)^a2
Development Time = b1 x (Effort)^b2

Last-minute checklist

- Understand the role of a project manager.
- Remember project-planning activities.
- Learn four estimation methods.
- Practise LOC formula and solved example.
- Learn five Function Point components and FP formula.
- Remember COCOMO formulas, KLOC, and project types.
- Learn scheduling principles and 40-20-40 rule.
- Understand Gantt chart, PERT, CPM, and critical path.
- Learn risk types, probability, impact, RMMM, and RIS.
- Revise the quick revision table and formulas.

Final note: If you understand the role of the project manager, estimation formulas, scheduling methods, and risk management, you can answer most questions from Unit 5.